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RESEARCH ARTICLE

Temporal-Contextual Attention Network for Solid-State Drive Failure Prediction in Data Centers

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ABSTRACT Proactive strategies for predicting solid state drive(SSD) failures are imperative to ensure uninterrupted services in data centers. Traditional methods that rely on rule-based approaches and machine learning algorithms often fail to accurately predict these failures. In this study, we introduce a temporal-contextual attention network(TCAN), a pioneering method that integrates long short-term memory(LSTM) and transformer architectures to address the limitations of existing schemes. TCAN exploits temporal patterns and attribute dependencies and offers a more comprehensive solution for SSD failure prediction. Unlike conventional feature selection methods, TCAN adopts a feature grouping approach that utilizes all available attributes while accounting for its unique characteristics. Specifically, TCAN treats certain features based on their temporal aspects, capturing how these features change over time, whereas other features are used to capture the dependencies and interactions among different attributes. Through extensive evaluation of private datasets from the Tencent data center and comparison with state-of-the-art models, including machine learning and deep learning approaches, TCAN demonstrates superior performance in identifying potential failures. Furthermore, ablation studies and evaluations using public datasets validate their effectiveness and robustness across different datasets. Our findings underscore the importance of considering both temporal features and inter-attribute dependencies for accurate SSD failure prediction, highlighting the potential of TCAN for enhancing storage system reliability and service stability in data center environments.

INDEX TERMS Cross attention, data center, deep learning, disk failure prediction, feature grouping, SSD SMART data, time series classification.

I. INTRODUCTION

In recent years, solid-state drives(SSDs) have largely replaced traditional hard disk drives(HDDs) in the data storage domain. SSDs are preferred because of their superior performance, lower power consumption, and enhanced reliability. This paradigm shift extends beyond individual computing systems to data-intensive environments such as data centers and servers, where rapid processing and efficient management of large-scale data sets are crucial. In data centers, the deployment of SSDs has enabled the seamless handling of massive amount of data and complex computations.

With the increasing use of SSD, the need for proactive strategies to predict potential failures has become increasingly evident. Data center operators emphasize on predicting SSD device failures to prevent data loss, minimize service unavailability, and reduce maintenance costs. Despite redundant protection schemes such as RAID(Redundant Array of Independent Disks) and replication, recovering from service disruptions in data centers is costly and time-consuming, leading to significant financial losses [1].

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Proactively identifying faults offers numerous benefits, making them a critical focus for efficient data center operations [2], [3].

A widely employed scheme for predicting failure in the storage domain is the utilization of SMART(Self-Monitoring, Analysis and Reporting Technology), which encompasses various attributes that span SSD functionality, including hardware health information, performance (host/NAND flash, read/write) metrics, temperature profiles, and SSD operation-related features. Initially much of the research on SSD failure prediction focused on applying rule-based replacement policies on SMART where the SMART attributes preceding the predetermined thresholds indicated potential issues. However, these rigid approaches often led to premature drive replacements, owing to overly conservative rule sets. To overcome these shortcomings, there has been a shift toward more flexible and accurate methodologies using machine learning and deep learning models. These advanced approaches aim to provide better predictions and reduce unnecessary drive replacements.

However, existing models still have several limitations, prompting the need for innovative approaches. First, SSDs exhibit unique characteristics: Unlike HDDs which consist of mechanical parts where hardware-induced errors occur more instantaneously, SSDs exhibit gradual changes toward failure over time, with multiple attributes contributing to this process. Consequently, capturing both the temporal features and attribute relationships is of paramount importance. Second, there is a discrepancy between the intentions of SSD manufacturers and their research priorities, particularly regarding feature selection. Although manufacturers aim to enhance monitoring and failure prediction by introducing additional attributes, current research often emphasizes excessive feature minimization, leading to suboptimal predictive models.

Motivated by these challenges, this study introduces a novel deep learning-based model called the temporal-contextual attention network(TCAN). Our model seeks to leverage the strengths of both long short-term memory (LSTM) and transformer architectures by integrating cross-attention mechanisms to address the complex nature of SSD failure prediction. Previous studies have often employed a serial approach - using transformers or CNN followed by LSTM networks. Although simpler to implement and train, such approaches may not capture the sophisticated interactions between temporal and contextual information in data. By employing LSTM-generated queries within a transformer framework, our model excels at detecting subtle interactions between temporal and contextual information, enhancing its ability to predict SSD failures accurately. To further address the limitations of prior methodologies, we propose a feature grouping approach that leverages all available attributes and utilizes each attribute according to its trends. In doing so, TCAN aligns with industry preferences and considers the intrinsic characteristics of each feature while achieving superior performance in failure identification. This innovative approach bridges the gap between industry demand and the limitations of previous research methodologies. This study makes two main contributions:

1. We formulated a novel disk failure prediction model that integrates the long short-term memory (LSTM) and transformer architectures using cross-attention mechanisms. This model leverages the strengths of both the transformer and LSTM networks, effectively capturing temporal features and dependencies among attributes. This is crucial for predicting failures given the nature of SSDs.
2. Unlike previous studies that primarily focused on feature selection, we propose a methodology that utilizes all available attributes without extensive feature reduction. By adopting a feature grouping approach, we can utilize different features based on their characteristics. Some features require a time series approach to track their progression over time, whereas others require capturing the dependencies and interactions among different attributes. We demonstrate the effectiveness of this feature grouping by showing how to utilize each group according to its specific usage while maximizing the utility of additional attributes introduced by SSD vendors in predicting SSD failures.

II. RELATED WORK

Research on drive failure prediction spans various methodologies tailored to both HDDs and SSDs. Traditional methods that rely on statistical approaches, such as the Mahalanobis distance [4], Wilcoxon rank-sum test [5], and Gaussian mixture method [6], offer binary predictions of drive failure but may struggle to capture gradual changes in drive health. With more recent advancements, machine learning-based techniques have gained prominence in drive failure prediction. For HDDs, binary failure prediction utilizing Bayesian approaches [7], hidden Markov models [8], support vector machines (SVM) [9], and classification tree models have been explored. Similarly, deep learning methods that utilize temporal features, such as recurrent neural networks (RNNs) [10], [11] and LSTM [12], have shown promise in predicting HDD failures.

With the rise in SSDs and their distinct characteristics, many failure prediction methods have emerged. Some studies have employed supervised learning classification algorithms, such as logistic regression, k-NN, SVM, and random forest to predict SSD failures. Various ensemble models have been used. In [13] various classification models were explored and the superior performances of tree ensemble models, specifically XGBoost, were demonstrated. In [14], a novel approach called part-voting random forest for predicting drive failure was introduced. Stacking of ensemble models including XGboost and LSTM, was proposed in [15], and in [16], multi-view and multi-task random forest(MVTRF), an ensemble of random forests that incorporates input beyond raw features, including histogram and sequence-related features, was proposed.

Furthermore, recognizing the success of transformer models in various domains, considerable efforts have been made to apply them to drive failure prediction tasks. Models such as NTAM [17] explore the influence of neighboring disks and incorporate neighborhood information along with temporal analysis, whereas ConvTrans-TPS [18] employs a convolutional transformer to apply convolutional projection for query, key, value to focus on local context and enhance performance. Efforts to integrate various methodologies have been pivotal for advancing the fusion of different methods. In [19], a temporal CNN(TCNN) model that is superior to RNN and LSTM was proposed. The integration of CNN and LSTM was proposed in [20] so that the CNN-LSTM model can not only select better features but also effectively learn sequential data. In [21], a variational autoencoder(VAE) was successfully integrated with LSTM. These models successfully extracted diverse features using different models. In addition, unsupervised anomaly detection algorithms, including 1-class isolation forest and autoencoders, were introduced [22], which have also been utilized to predict SSD failures by learning patterns only from healthy SSDs and identifying anomalies through reconstruction errors when exceeding the threshold. Moreover, there has been growing interest in integrating real-time data with historical data, akin to digital twin-driven fault diagnosis methods. Digital twins, which combine virtual and real data for fault diagnosis, have proven effective for composite fault detection in various systems [23], [24], [25].

In addition to exploring models and algorithms, effective feature selection schemes have been investigated. Xu et al. [26] introduced the Wearout-updating Ensemble Feature Ranking(WEFR) to select the SSD SMART features with high discriminative power. Lu et al. [20] applied the J-index to select learning features including SMART attributes, performance metrics, and locations for HDD failure prediction. Furthermore, in [17] and [18] the temporal prediction synthesis(TPS) method was introduced, which is an effective data augmentation approach for mitigating extreme data imbalance issues and enhancing the robustness of SSD failure prediction models.

III. PRELIMINARIES

This section explains the fundamental concepts of the transformer and LSTM architectures, which are essential components of the proposed drive failure prediction models. Furthermore, the key distinctions between self-attention and cross-attention mechanisms, which are crucial elements within the transformer architecture, are identified. Finally, we provide the motivation for integrating these models.

A. TRANSFORMER

Transformer architecture represents a significant advancement in deep learning primarily owing to the incorporation of self-attention mechanisms [27]. Widely applied in diverse fields such as computer vision and natural language processing, the transformer utilizes an encoder-decoder structure with key components such as feedforward layers and self-attention mechanisms. Through self-attention, the model efficiently captures the dependencies among features. Mathematically, attention calculation involves operations on the query, key, and value matrices. These operations allow the model to focus on the relevant information within the input features. This capability makes the transformer adept at identifying the relationships and dependencies within sequences or attributes.

While the transformer model boasts numerous strengths, its core operations, centered on query, key, and value calculations, do not inherently encode temporal order in sequential data. Although it incorporates positional embedding to address this limitation, it cannot naturally integrate the temporal aspects into mathematical formulations. This characteristic can present challenges in scenarios in which the sequence of events plays a critical role.

B. LONG SHORT-TERM MEMORY(LSTM)

LSTM is a specialized form of Recurrent Neural Network(RNN) designed to capture long-term dependencies in sequential data such as time series data. Unlike conventional RNNs, which struggle with vanishing or exploding gradients when learning long-range dependencies, LSTM incorporates memory cells, which are capable of retaining information over extended sequences. Each LSTM cell contains three key gates: the input gate, forget gate, and output gate. These gates regulate the flow of information into, within, and out of the cell to help the network remember information over long periods. This ability is vital for capturing temporal dependencies inherent in time series data.

C. SELF ATTENTION AND CROSS ATTENTION IN TRANSFORMER

Self-attention allows each feature in a row to attend to all other features, thereby capturing interdependencies and contextual relationships. By utilizing the learned attention weights, each feature computes a weighted sum of all the features, thereby enhancing its representation based on the relevance of other features. This approach enables models to effectively capture complex patterns and dependencies, potentially improving their performance in tasks such as classification or regression.

By contrast, cross-attention is employed in scenarios where the lengths of query and key/value sequences vary, enabling the integration of information across distinct sequences or sources. Unlike self-attention, which focuses solely on the features within the same sequence, cross-attention facilitates information exchange between different sequences. This process involves using attention masks from one sequence to highlight relevant features extracted from another sequence. Extensive research in analogous domains, such as defect detection in magnetic tiles [28] and automotive components [29], has leveraged cross-attention to address similar challenges and demonstrated promising performance.

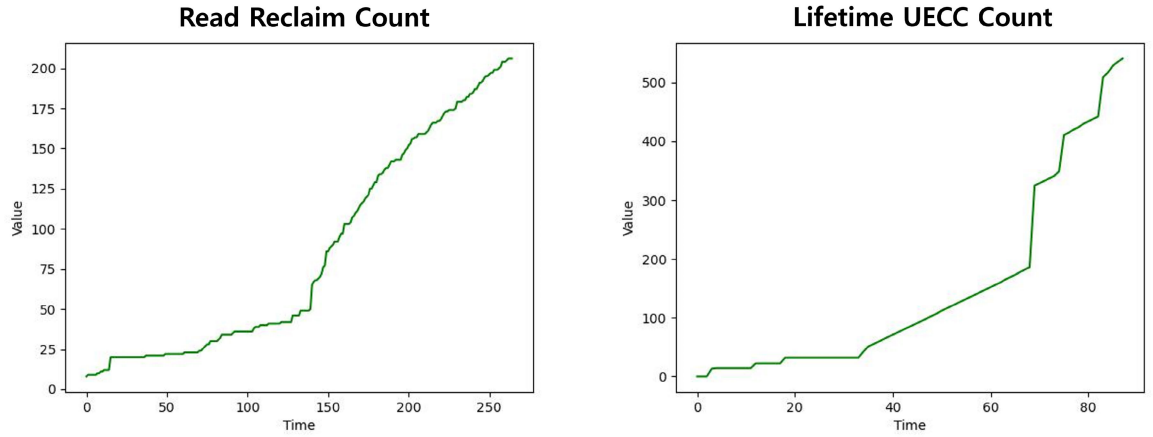


FIGURE 1. Examples of monotonic attributes.

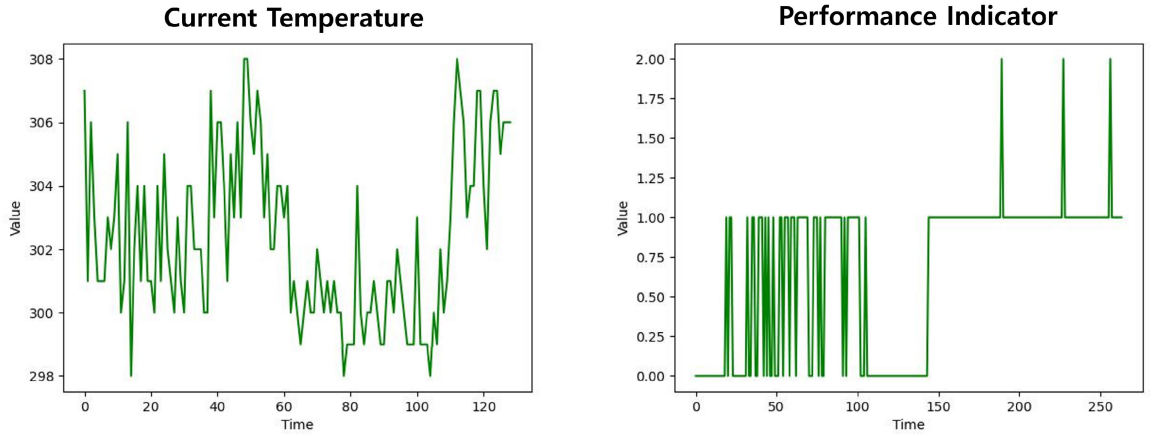


FIGURE 2. Examples of randomly fluctuating attributes.

IV. METHOD

In this section, we introduce our novel drive failure prediction model TCAN. We begin by outlining feature grouping, which is the process of grouping the SSD SMART attributes based on their unique characteristics. Subsequently, we examine the architecture of the model to provide insights into its robust predictive capability.

A. FEATURE GROUPING

Although much of the existing research on SSD failure prediction highlights the necessity of feature selection, there is a growing demand for streamlining the burden of feature engineering. In response to this industry's needs and the increasing diversity of attributes introduced by SSD vendors, we propose a simple yet powerful technique called feature grouping.

SSD SMART information typically comprises a multitude of attributes, each offering insights into the health and performance of a drive. Attributes can be broadly categorized into two types based on their characteristics and contribution to the prediction of SSD failures.

1) MONOTONIC ATTRIBUTES

The majority of SMART attributes fall into the category of monotonic behavior, monotonic attributes exhibit consistent, predictable behavior over time, indicating a gradual progression towards failure, which aligns with SSD failures that often result from gradual wear and changes. Examples of such attributes include the end-to-end error count, program/erase fail counts, and uncorrectable error count as shown in Fig. 1, which provide valuable insights into the long-term health trajectory of SSD by capturing gradual changes that are indicative of impending failure or degradation.

2) RANDOM FLUCTUATING ATTRIBUTES

These attributes introduce variability and unpredictability into the monitoring process. These values typically monitor external environmental factors and have less direct temporal relevance to SSD operations. Although their patterns may lack clear trends, they still offer valuable information about the operational status of an SSD. Examples of such attributes include temperature fluctuations, volatile error counts,

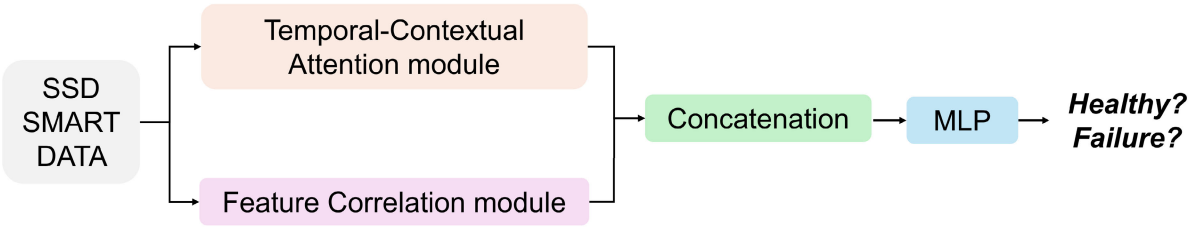


FIGURE 3. Overall architecture of temporal-contextual attention network.

and error addresses such as DRAM CECC addresses as shown in Fig. 2. Despite their erratic nature, these attributes contribute to the overall understanding of SSD health and can serve as early indicators of potential health issues.

The relationship between monotonic and random fluctuating attributes are as follows: M represents monotonic feature vectors, and R represents random fluctuating feature vectors, where $f_i(j)$ represents the j-th features of the i-th data point.

$$\begin{aligned} M &= \{j \mid f_i(j) \text{ is a monotonic feature}, \\ &\quad j \in \{1, 3, 4, \ldots, j\}\} \\ R &= \{k \mid f_i(k) \text{ is a random fluctuation feature}, \\ &\quad k \in \{2, 5, \ldots, k\}\} \\ M \cup R &= \{1, 2, \ldots, n\} \text{ and } M \cap R = \emptyset \end{aligned}$$

For monotonic attributes, a time series approach is necessary to analyze how specific metrics evolve over time and recognize the patterns and trends that precede failure. Conversely, the temporal features are less significant for randomly fluctuating attributes. However, understanding their interactions, such as the relationship between the temperature and error rates, can enhance the performance of failure predictions.

B. MODEL ARCHITECTURE

Our TCAN model, considers these two categories of features: monotonic and random fluctuations, differently. It leverages feature grouping and cross-attention mechanisms for harnessing both the attribute dependencies and temporal characteristics. By identifying feature combinations associated with defects and capturing the temporal characteristics of steady, cumulative counts, TCAN aims to provide a comprehensive understanding of defect-related feature patterns. The overall architecture of TCAN is shown in Fig. 3. The model comprises two primary modules: temporal-contextual attention(TCA) and feature correlation(FC). The TCA module captures the temporal progression of monotonic features and the intricate relationships among randomly fluctuating features. The FC module effectively captures global feature dependencies by initially adding a learnable CLS token. Finally, both temporal and contextual features from these two distinct modules are concatenated and processed using a simple multi-layer perceptron(MLP) to predict the status of the SSDs.

Both the modules share the same input, denoted by x_i. Input $\mathbf{x}_i$ refers to feature vectors with i number of attributes. Each feature vector $\mathbf{f}_i$ has n sequence length. The target variable y_i is binary and labeled as either 0 or 1.

$$\mathbf{x}_i = [f_1, f_2, \ldots, f_i] \tag{1}$$

$$\mathbf{f}_i = [f_i(1); f_i(2); \ldots; f_i(n)] \tag{2}$$

1) TEMPORAL-CONTEXTUAL ATTENTION MODULE

The TCA module is designed to capture the temporal progression of monotonic features and intricate relationships among randomly fluctuating features. This module employs both self-attention and cross-attention mechanisms, as outlined in (4). As shown in Fig. 4, monotonic feature vectors are processed through a global LSTM to extract their temporal characteristics by utilizing cross-attention to focus on relevant features. Specifically, the query matrix is derived from the output of the global LSTM, whereas the key and value matrices are derived from the monotonic feature vectors, where H represents the hidden state matrix from LSTM, and W_Q, W_K, W_V are all learnable matrices, as shown in (3). Through this cross-attention mechanism, the global temporal feature acts as a query, highlighting relevant features extracted from the key and value matrices.

Concurrently, randomly fluctuating attributes pass through a simple Transformer encoder to perform self-attention, where the Query, Key, and Value matrices are derived from the same random fluctuation feature vectors. This process discerns relationships within the fluctuating features, revealing dependencies that may indicate potential defects.

Finally, the outputs from both the monotonic feature encoder and the random fluctuation feature encoder are concatenated and processed through an LSTM. Here, f_t represents the feature at time step t, and h_t represents the hidden state of LSTM at time step t, as shown in (5). This concatenation allows the model to combine the extracted temporal features and the learned dependencies from both types of attributes, providing a comprehensive understanding of the SSD's health status.

$$\begin{aligned} Q = W_Q \cdot H,\ K = W_K \cdot M, \\ V = W_V \cdot M \end{aligned} \tag{3}$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{4}$$

$$h_t = \text{LSTM}\,(f_t, h_{t-1}) \tag{5}$$

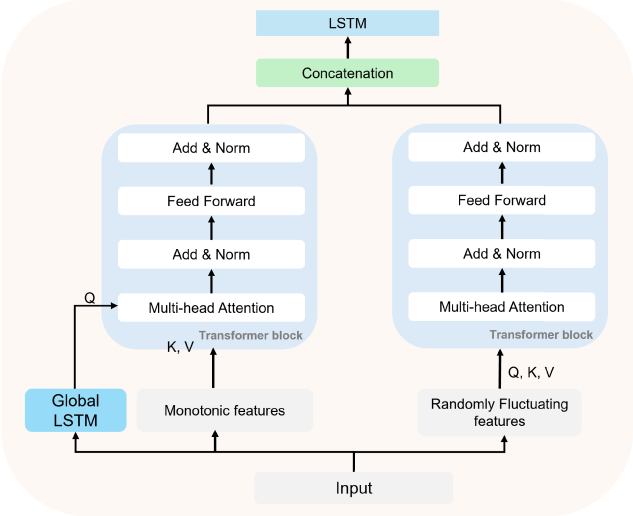


FIGURE 4. Architecture of temporal-contextual attention module in temporal-contextual attention network.

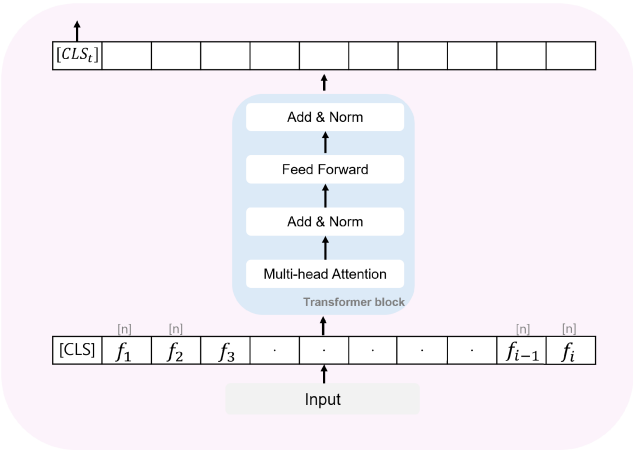


FIGURE 5. Architecture of feature correlation module in temporal-contextual attention network.

2) FEATURE CORRELATION(FC) MODULE

Understanding the interactions between attributes in terms of failure, such as the relationship between performance and temperature, can reveal important insights into the failure mechanisms. Inspired by BERT-like architectures [30], the FC module depicted in Fig. 5 incorporates a trainable embedding token [CLS] at the beginning of the input feature vector x_i. This token facilitates the learning of inter-attribute correlations. In conventional transformer architectures, self-attention tends to emphasize the feature itself, potentially neglecting inter-attribute dependencies. However, the inclusion of the CLS token enables the model to learn the dependencies within attributes effectively. Thus, the original input $\mathbf{x}_i$ becomes $\mathbf{x'}_i$ as shown in (6), which is a single data point with features f_i (i is the index of the attributes, and each f_i has n sequence length) and is fed into to the transformer block. As a result, the classification token CLS_t encapsulates the global contextual relationships.

TABLE 1. 71 SMART attributes.

Attribute Names	*Attribute Names*
program fail count	sram cecc address coresram
erase fail count	sram cecc count buffer
wear leveling min	sram cecc address buffer
wear leveling max	dram cecc count
wear leveling avg	dram cecc address
end to end error detection count	e2e error count
crc error count	fw update success count
timed workload media wear 0xca	fw update fail count
timed workload host reads	highest temperature 0xca
timed workload timer	lowest temperature 0xca
thermal throttle status cnt	read recovery attempts 0xca
nand bytes written	reset count
host bytes written	trimmed sector count
lifetime WAF	over temperature count
trailing hour WAF	recovery reset count
lifetime user writes	throughput performance
lifetime nand writes	timed media wear 0xc4
lifetime user reads	timed host read write ratio
lifetime retired block count	read error rate
current temperature	read recovery attempts 0xc4
capacitor health	highest temperature 0xc4
lifetime unused reserved block	lowest temperature 0xc4
read reclaim count	physical media units written
lifetime uecc count	physical media units read
lifetime used reserved block	bad user nand block count
power on hours	bad system nand block count
total normal power off count	E2E correction count detected
total sudden power off count	E2E correction count corrected
perf indicator	system data used
wearlevel count	free blocks
hw error type	unaligned io
devicefail recovery count	security version number
sram cecc count coresram	namespace utilization
bad tlp count	plp start count
bad dllp count	endurance estimate
phy error count	

$$\mathbf{x'}_i = [\mathrm{CLS}, f_1, f_2, \ldots, f_i] \tag{6}$$

$$\mathrm{CLS}_t = \mathrm{Transformer}\left(\mathbf{x'}_i\right) \tag{7}$$

$$y = \mathrm{MLP}\left([\mathrm{CLS}_t; h_T]\right) \tag{8}$$

The last part of the model is a concatenation of the outputs from the two modules and the classification, as outlined in (8). $[\mathrm{CLS}_t; h_T]$ represents the concatenated output from the FC module and the final hidden state from the TCA module, which are fed into a MLP for classification. The model's output is designed to predict the probability of disk failure, with outputs greater than 0.5 classified as failed drives and outputs less than 0.5 classified as healthy drives.

V. DATA AND EXPERIMENT

In this section, we provide a comprehensive overview of the dataset utilized in our study, including its characteristics and relevance to our research objectives. In addition, we present a streamlined approach that covers experimental setup, preprocessing procedures, and deployment framework. Finally, we present the evaluation metrics employed to assess the performance of our method.

A. DATA OVERVIEW

We assessed TCAN using industrial private datasets obtained from real-world data centers, which comprise two main components: Offline and Online datasets.

The **online dataset** consists of daily SMART logs obtained from the SSDs deployed within the data centers at Tencent Cloud, specifically focusing on Samsung PM9A3 SSDs. This dataset spanned a duration of one year, from January 2022 to December 2022. It comprised data from 181,507 SSDs, of which 181,132 were deemed healthy and 375 experienced failures. The SMART logs contained a multitude of attributes, including SSD and server serial numbers, timestamps, and vendor-specific attributes, which are collected daily using vendor tools as shown in Fig. 6. The dataset spanned 71 SMART attributes, showcasing a rich variety of data points for analysis. All 71 SMART attributes are provided in Table 1.

By contrast, the **offline dataset** consisted of data gathered from the monitoring system of the data center, which continually monitors anomalies or failures based on pre-defined rules. Alternatively, on-site engineers may conduct manual inspections for abnormal behaviors. Upon detection, relevant information regarding the failed SSD, such as its serial number, server serial number, and time of failure, is recorded in the abnormality reports. This dataset contained the records of 1,200 SSD failures.

To establish accurate labels for the SSD status (normal or failed), we matched the records from the online and offline datasets. In this process, the status of each SSD was categorized as either normal (0) or failure (1) based on the presence or absence of corresponding records in the offline dataset. If an SSD serial number was found in the abnormality reports of the offline dataset, it was labeled as a failure (1); otherwise, it was labeled as normal (0). This meticulous matching process ensured precise classification of the status of each SSD in the online dataset.

B. EXPERIMENTAL SETUP

To prevent data leakage due to temporal information, the datasets were split into training, validation, and test sets based on the unique index of the SSDs. The ratio of the splits was maintained at 6:2:2 for the training, validation, and test sets, respectively, based on the number of failures. The training set was used to train the model, while the validation set was used to fine-tune the hyperparameters of the model by evaluating its performance during the training iterations. Finally, a test set was used to conduct a conclusive assessment of model performance.

Throughout our evaluation process, we conducted three individual tests on different data subsets, and the final results were derived by averaging the outcomes of these independent experiments to ensure the robustness and reliability of our evaluation metrics. The TCAN model was trained by minimizing the binary cross-entropy(BCE) loss using the Adam optimizer, with the model parameters updated through back-propagation during training. $p(y_i)$ is the probability of a failed SSD, and $1 - p(y_i)$ is the probability of healthy SSD. The hyperparameters were set as follows: 0.0001 learning rate, four hidden layers in an MLP, eight heads in a multi-head attention mechanism, and a batch size of 64.

$$\text{BCE loss} = -\frac{1}{N}\sum_{i=1}^{N} \mathrm{y}_i \cdot \log\left(p\left(\mathrm{y}_i\right)\right) + \left(1 - \mathrm{y}_i\right) \cdot \log\left(1 - p\left(\mathrm{y}_i\right)\right) \tag{9}$$

TABLE 2. Summary of industrial private dataset.

Dataset	Number of Unique SSD	Number of Rows
Healthy Drive	187,772	27,170,388
Failed Drive	375	375

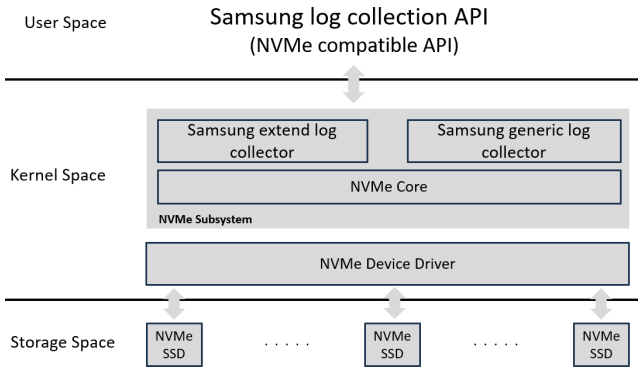


FIGURE 6. Flow chart illustrating the extraction of SMART logs from data center.

C. DATA PREPROCESSING

We transformed all collected data into a numeric format to ensure compatibility with our models, given the varied formats such as string, date-time, and float. During preprocessing, string values such as the drive model, firmware name, model name, and server serial number were dropped. Additionally, we normalized all data to fit within the range of 0-1 using(10). Here, $x_{\min}$ and $x_{\max}$ represent the minimum and maximum values of the training dataset, respectively. This normalization process ensured that the data are scaled appropriately to our deep learning models.

$$x_{\text{new}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{10}$$

To extract the temporal features, we organized each data point to have the same sequence length. This involved grouping specific drive serial numbers with varying timestamps to form a structured representation. Using the SSD failure timestamp as the reference point, we selected a sequence length of 10 to utilize the maximum number of failed SSDs and effectively capture the temporal characteristics associated with failure occurrences. The input data consists of multiple features recorded across a sequence of time steps, represented as a tensor with the shape (number of attributes * sequence length). Each attribute corresponds to SSD SMART attributes, while the sequence length denotes

the number of time steps or historical data windows used for the prediction. Additionally, to better capture information regarding abnormalities and mitigate noise in the normal data, we undersampled the normal data.

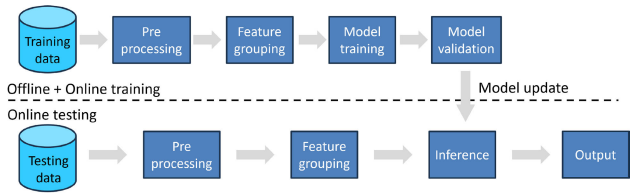


FIGURE 7. Block diagram of deployment framework of TCAN.

D. DEPLOYMENT FRAMEWORK OF TCAN IN PRACTICAL APPLICATIONS

As shown in Fig.7, the deployment framework consists of two main phases: training/validation and testing. During the training phase, the data were trained using the online SMART and offline failure information data, and validation was conducted to determine the best parameters for the model. Testing was conducted online during the inference stage using validated parameters.

Specifically, the process begins with the utilization of offline failure information and online SMART logs to classify the status of each SSD as normal or failing. The SMART logs are then organized into time series data, discarding noisy or incomplete data points. Based on the trends of the SMART attributes, they are categorized into two groups: monotonic and randomly fluctuating. This preprocessed dataset forms the foundation for training, validation, and testing.

E. METRICS

In evaluating the performance of our SSD failure prediction models, we prioritized Precision, Recall, F1-Score, and AUC_ROC, as these metrics are more suitable for imbalanced datasets and better aligned with the operational needs of data centers, where failure events are rare. These metrics offer a more accurate assessment of the model's ability to detect critical failures while minimizing false positives and false negatives. Given that a false alarm, such as replacing an SSD based on a false positive, incurs minimal cost compared to the significant downtime and operational losses caused by missing an actual failure, these metrics provide a more meaningful evaluation than accuracy. While accuracy is commonly used in many machine learning applications, it is not emphasized here due to its limitations in imbalanced datasets, where high accuracy can be achieved by simply predicting the majority class. However, for completeness, we also report the accuracy of each model.

1) PRECISION

This metric assesses the ratio of correctly predicted failed SSDs (true alarms) to all predicted failed SSDs, including true and false alarms.

2) RECALL

Recall, also known as the true positive rate (TPR), measures the proportion of correctly predicted failed SSDs to actual failed SSDs.

3) F1-SCORE

F1-Score represents the harmonic average of precision and recall, offering a balanced assessment of the model's capability to correctly detect failed SSDs while minimizing false alarms.

4) AUC_ROC

AUC_ROC, short for area under receiver operating characteristic curve, quantifies how effectively the model distinguishes between two classes (normal and failed) in the dataset. Plotting the TPR against the false positive rate(FPR), provides a single score indicative of the diagnostic capability of the binary classification model. The AUC_ROC score ranges from 0.5 to 1.0, where 0.5 signifies a performance equivalent to that of random guessing. Given our highly imbalanced dataset, this metric is particularly useful because it remains robust and insensitive to class imbalance.

$$\begin{aligned}
\text{Precision} &= \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \\
\text{Recall(TPR)} &= \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \\
\text{FPR} &= \frac{\text{False Positives}}{\text{False Positives} + \text{True Negatives}} \\
\text{F1-Score} &= \frac{2 \times \text{ Precision } \times \text{ Recall}}{\text{Precision } + \text{ Recall}} \\
\text{AUC_ROC} &= \int_0^1 TPR\left(FPR^{-1}(x)\right) dx
\end{aligned} \tag{11}$$

Here, $FPR^{-1}(x)$ is the inverse function of FPR with respect to the threshold x, and x ranges from 0 to 1.

VI. EVALUATION

In this section, we present a comprehensive evaluation of TCAN using both private and public datasets. In addition, we discuss the ablation studies performed to evaluate the contribution of each module and feature grouping to performance.

A. RESULTS AND ANALYSIS OF TCAN MODEL

1) FAILURE PREDICTION PERFORMANCE

We assessed the performance of TCAN by comparing it with various machine learning and deep learning-based approaches for drive failure prediction. These include well-established machine learning algorithms such as **SVM**, **random forest** with a maximum depth of 5(**RF_5**) and 10(**RF_10**), and a **light gradient boosting machine** with DART(**LGB_D**) and GBDT(**LGB_G**) boosting methods. These algorithms are widely used for drive failure prediction tasks. In addition, we evaluated several deep learning-based methods, including **autoencoder(AE)**, **stacked**

TABLE 3. Parameter configuration of different models.

	Kernel Type	Max depth of a tree	Number of Layers	Activation Function	Learning Rate	Batch Size	Dropout Rate
SVM	RBF	-	-	-	-	-	-
Random Forest	-	5	-	-	-	-	-
LightGBM	-	12	-	-	-	-	-
Autoencoder	-	-	4	ReLu	0.001	64	0.2
Neural Network	-	-	4	ReLu	0.001	64	0.2
LSTM	-	-	3	ReLu	0.001	64	0.2
Transformer	-	-	4	ReLu	0.001	64	0.1
TCAN	-	-	4	ReLu	0.001	64	0.1

TABLE 4. Performance of TCAN and nine other models on the industrial private dataset.

Model	*Precision*	*Recall*	*F1-Score*	*AUC*
SVM	0.63	0.50	0.56	0.58
RF_5	**0.75**	0.57	0.65	0.57
RF_10	0.73	0.62	0.67	0.62
LGB_G	0.73	0.65	0.69	0.66
LGB_D	0.71	0.64	0.67	0.64
AE	0.58	0.31	0.40	0.53
NN	0.55	0.52	0.54	0.61
LSTM	0.68	0.66	0.67	0.69
TF	0.55	0.61	0.58	0.63
TCAN	0.73	**0.68**	**0.70**	**0.73**

LSTM(LSTM), **neural network(NN)**, **transformer(TF)**. These deep learning models have been widely adopted and have demonstrated remarkable performance in predicting drive failures. Throughout the evaluation, we conducted 5-fold cross-validation to obtain a more reliable estimate of the model's performance while maximizing the use of available failure data. Table 3 is the specifics of the parameter tuning for each model used in our comparison. Each model was trained and evaluated using the best parameter configuration identified through Grid Search. We conducted multiple trials for each configuration to ensure the robustness and reliability of the results. This approach guarantees that the reported performance reflects the models' optimal settings. Although accuracy is not our primary evaluation metric, given its limitations in the context of imbalanced datasets and data centers' requirement, we report it here for reference, as it may be of interest to some researchers. The accuracy for each model is as follows: SVM (87.4%), Random Forest (97.2%), LightGBM (98.1%), Autoencoder (80.1%), LSTM (91.1%), Transformer (95.2%), and TCAN (97.9%).

Table 4 presents the results of these methods and TCAN. Values highlighted in bold indicate the best performance. The observations reveal that tree ensemble models, including RF and LGB, generally exhibited higher precision. An AE that learns the features of healthy drives and judges health/failure based on a reconstruction loss threshold, requires meticulous threshold control. However, based on the results, it showed the lowest recall and F1-score, indicating its inability to distinguish between the characteristics of healthy and failed drives within the dataset. The TF showed an average F1-score and AUC, whereas LSTM achieved an average recall of 0.66 and outperformed previous methods. In terms of metrics, TCAN enhanced precision by 7.1%, recall by 15.3%, F1-score by 9.4%, and ROC_AUC by 10.6% on average compared with the existing approaches.

TCAN significantly outperformed other approaches, and LSTM also demonstrated a relatively high performance compared with the other models. This can be attributed to the nature of the SSD SMART attributes, which inherently contain information showing gradual changes toward failure rather than instant failure, thus highlighting the importance of capturing not only contextual but also temporal features in SSD failure prediction tasks.

2) ABLATION EXPERIMENT

We conducted additional experiments to demonstrate the effectiveness of TCA and feature grouping mechanisms in TCAN. To explore these aspects further, we modified TCAN as follows:

- TCAN_ver1: TCAN was modified such that the TCA module performs self-attention instead of cross-attention by removing the global LSTM.
- TCAN_ver2: The FC module was disabled to assess its contribution to performance.
- TCAN_ver3: All SMART attributes were combined into a single group without feature grouping and TCA was performed.
- TCAN_ver4: The position of the input groups was switched between monotonic and random fluctuating attributes to analyze how each group interacts with the temporal features.

The results of the ablation experiments are summarized in Table 5. The performance of TCAN_ver1 demonstrates the efficacy of TCA, underscoring the importance of both temporal and contextual dependencies in distinguishing failures from normal occurrences. Furthermore, TCAN_ver2 demonstrates that disabling the FC module negatively affects the performance of TCAN, indicating that concatenation with the FC module enhances the effectiveness of TCAN beyond what the TCA module alone achieves. Subsequently, TCAN_ver3 and TCAN_ver4 shed light on feature grouping and the significance of monotonic attributes when incorporating temporal aspects. TCAN_ver3 combines all SMART attributes into a single group without feature grouping, showcasing the effectiveness of feature grouping

TABLE 5. Ablation - TCAN and its variants.

Model	*Precision*	*Recall*	*F1-Score*	*AUC*
TCAN	0.73	**0.68**	**0.70**	**0.73**
TCAN_ver1	0.59	0.63	0.61	0.63
TCAN_ver2	0.68	0.63	0.65	0.68
TCAN_ver3	**0.74**	0.65	0.69	0.71
TCAN_ver4	0.71	0.59	0.65	0.69

TABLE 6. Summary of Backblaze dataset.

Dataset	Number of Unique SSD	Number of Rows
Healthy Drive	42,449	14,759,143
Failed Drive	906	906

in improving the model performance. Conversely, the modification of TCAN_ver4, which switches the position of the input groups between monotonic and random fluctuating attributes, emphasizes the necessity for monotonic attributes to incorporate temporal aspects. Although TCAN_ver3 and TCAN showed little variance in performance, it can be inferred that TCA plays a more pivotal role in distinguishing failures from normal occurrences than feature grouping.

B. EVALUATION ON PUBLIC DATASET

We conducted further empirical analyses to assess the performance of TCAN and feature grouping using the public Backblaze dataset [31]. This dataset comprised various HDD models from different vendors spanning a significant duration and data were well-labeled with minimal missing values or outliers. This has been widely utilized in previous studies on drive failure prediction [19], [32], [33], [34], [35].

The Backblaze dataset included the drive model, serial number, timestamp, SMART attributes, and labels (normal or failed) for each drive. For our analysis, we utilized the most recent data from January 2023 to December 2023. From the dataset, we selected three models from Seagate (ST4000DM000, ST12000NM001G, and ST14000NM001G) because drives from the same vendor share similar or identical SMART attributes and these models exhibit a moderate number of healthy and failed drives. The model ST4000DM000 contained 44 SMART attributes, while ST12000NM001G and ST14000NM001G had 38 attributes. To ensure consistency, we excluded six attributes from ST4000DM000 to align them with the other two models. The dataset consisted of 42,449 healthy drives and 906 failed drives, with 14,759,143 rows for healthy drives and 906 rows for failed drives. On average, approximately 348 days of data were available for each unique drive.

To ensure the independence and uniqueness of our analysis, we divided the dataset equally into training, validation, and testing sets.

These findings suggest that TCAN outperformed AE, NN, and other deep learning methods in terms of F1-score and AUC. However, tree-based ensemble algorithms, particularly LGB, consistently demonstrated above-average performance, particularly in terms of AUC. While TCAN exhibited notably strong performance on the Backblaze dataset, with a 13% higher F1-score and an 8% higher AUC than the average, it closely rivaled the top-performing models overall. This contrasts with its performance on the SSD dataset, where TCAN outperformed all other models.

TABLE 7. Performance of TCAN and existing models on the Backblaze dataset.

Model	*Precision*	*Recall*	*F1-Score*	*AUC*
SVM	0.65	0.25	0.63	0.62
RF_5	**0.70**	0.53	0.60	0.76
RF_10	0.61	0.7	0.65	0.82
LGB_G	0.28	0.85	0.42	0.87
LGB_D	0.27	**0.89**	0.41	**0.88**
AE	0.54	0.33	0.41	0.71
NN	0.28	0.61	0.39	0.68
LSTM	0.48	0.67	0.56	0.73
TF	0.55	0.54	0.54	0.76
TCAN	0.65	0.68	**0.66**	0.85

This discrepancy in performance can be attributed to the emphasis of TCAN on both temporal features and feature dependencies. Notably, the Backblaze dataset comprising HDD logs appears to be straightforward and easily identifiable using boosting mechanisms. This may be because failures are not solely dependent on long-term time series characteristics but rather on sudden increases in specific SMART attributes. Because HDDs are composed of mechanical parts, hardware-induced errors are more intuitive. Therefore, capturing temporal features may be considered less crucial. This intuition is evidenced by attributes like SMART 5(Reallocated Sectors Count), SMART 10(Spin Retry Count), SMART 188(Command Timeout), SMART 197(Current Pending Sector Count), SMART 198(Uncorrectable Sector Count), and SMART 240(Head Flying Hours), either individually or in combination, indicating signs of impending failures or a high chance failure.

By contrast, the private dataset used in this study involved SSDs made of NAND flash memory. Flash memory typically does not exhibit many instant failures such as sudden breakage of memory cells, and some failures are corrected using firmware recovery algorithms. Memory cells often gradually degrade, making failure prediction more challenging. TCAN's ability to handle both temporal and contextual features makes it better suited for such scenarios, where failures are often influenced by a combination of temporal and contextual factors, as is typical in SSDs. Note that LSTM, with its ability to process long-term sequences and capture long-term failure symptoms, also showed superior performance with respect to the private SSD dataset. However, it exhibited lower performance on the Backblaze HDD dataset compared with the other models. This contrast highlights the differences in the characteristics of HDDs and SSDs. Consequently, time-series models such as TCAN are better suited for SSD datasets.

VII. CONCLUSION

We introduced TCAN, designed to accurately predict SSD failures, a major concern in the industry for maintaining the

uninterrupted operation of storage systems and the stability of online services in data centers. TCAN is an innovative approach that blends LSTM and transformer architectures using cross-attention mechanisms. This fusion enables the capturing of both the temporal patterns of the SSD status and the interdependencies among the SMART attributes regarding failure.

This unique approach not only addresses the specific needs of SSD failure prediction but also stands out from existing models that primarily focus on feature selection to minimize noise. Our model employs a targeted feature grouping strategy, differentiating between monotonic and fluctuating attributes, and optimally integrates these features into the prediction process. This method strategically leverages each feature based on its unique characteristics and contribution to failure prediction without extensive feature engineering, ensuring a robust and efficient model that aligns with industry standards. Our experimental results validate that feature grouping significantly enhances the SSD failure prediction performance.

We investigated the daily SMART logs for both SSDs and HDDs. More than 180,000 SSDs were gathered from Tencent data centers, and 43,000 HDDs were observed at the Backblaze data centers. SSDs and HDDs exhibit distinct characteristics: HDDs comprise mechanical parts and show signs of instant failure, whereas SSDs typically show gradual changes over time. Thus, capturing the temporal features is crucial for SSD datasets. Our experiments validated this hypothesis, demonstrating the superiority of TCAN over the other models, particularly for SSD datasets. TCAN consistently outperformed its competitors on the SSD datasets and demonstrated superior performance on the HDD datasets. The performance of LSTM on both the SSD and HDD datasets further bolsters our findings.

Overall, our results underscore the effectiveness of TCAN in addressing the specific challenges of SSD failure prediction and highlight its potential for improving reliability in storage systems.

In subsequent studies, we aim to gather additional SSD datasets from various models to validate the robustness and effectiveness of TCAN across different SSD datasets.

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